

Target leakage check: FatFreeW as a predictor of PerBFat1

Why this file exists

The main analysis (Multiple linear regression - Body Fat Index.qmd) retains FatFreeW (fat-free mass) in the final model. FatFreeW is not an independent body measurement. It is derived from the outcome:

$$\text{FatFreeW} = \text{Weight} \times \left(1 - \frac{\text{PerBFat1}}{100}\right)$$

so together with Weight, which is also in the dataset, it reconstructs the outcome almost exactly. Any model containing it reports a fit that cannot be achieved on a new man with a tape measure, which is the situation the study is for.

Density and PerBFat2 were already excluded at the start of the main analysis for exactly this reason. FatFreeW belongs in that same exclusion list.

This file quantifies the leak and refits the model without it.

```
d0 <- read.table("../data/fatdat.txt")
names(d0) <- c("ID", "PerBFat1", "PerBFat2", "Density", "Age", "Weight", "Height",
               "AdipInd", "FatFreeW", "Neck", "Chest", "Abdomen", "Hip", "Thigh",
               "Knee", "Ankle", "Biceps", "Forearm", "Wrist")
n <- nrow(d0)
```

1. The leak, demonstrated arithmetically

```
recon <- 100 * (1 - d0$FatFreeW / d0$Weight)
c(correlation = cor(recon, d0$PerBFat1),
  mean_abs_diff = mean(abs(recon - d0$PerBFat1)),
  max_abs_diff = max(abs(recon - d0$PerBFat1)))
```

```
correlation mean_abs_diff max_abs_diff
0.99562806 0.08696883 11.42756933
```

The correlation is 0.996 and the mean absolute difference is under 0.1 percentage points. The single large discrepancy is a known inconsistent record in the Penrose data, not a failure of the identity.

```
summary(lm(PerBFat1 ~ FatFreeW + Weight, d0))$r.squared
```

```
[1] 0.9580617
```

Those two variables alone explain about 96% of the outcome. That number is arithmetic, not prediction. Its cross-validated error is reported in section 5, after the helper below is defined.

2. Ten-fold cross-validation helper

```
cv_rmse <- function(formula, data, k = 10, seed = 1) {
  set.seed(seed)
  folds <- split(sample(nrow(data)), rep(1:k, length.out = nrow(data)))
  err <- unlist(lapply(folds, function(te) {
    fit <- lm(formula, data = data[-te, ])
    pred <- predict(fit, newdata = data[te, ])
    pred - data[[all.vars(formula)[1]]][te]
  }))
  sqrt(mean(err^2))
}
```

3. The model as originally reported

```
mod_reported <- lm(PerBFat1 ~ poly(FatFreeW, 2) + Height + Chest + Thigh + Wrist, d0)
c(R2 = summary(mod_reported)$r.squared,
  CV_RMSE = cv_rmse(PerBFat1 ~ poly(FatFreeW, 2) + Height + Chest + Thigh + Wrist, d0))
```

```
R2    CV_RMSE
0.8717133 3.8203140
```

4. The same selection with FatFreeW excluded from the candidate set

FatFreeW is removed before stepwise selection runs, not dropped afterwards. Dropping it at the end would already have let it shape the selection path.

```
d_clean <- subset(d0, select = -c(ID, PerBFat2, Density, FatFreeW))
mod_full <- lm(PerBFat1 ~ ., d_clean)
mod_clean <- step(mod_full, k = log(n), trace = 0) # BIC, explanatory goal
formula(mod_clean)
```

```
PerBFat1 ~ Weight + Abdomen + Forearm + Wrist
```

```
c(R2 = summary(mod_clean)$r.squared,
  CV_RMSE = cv_rmse(formula(mod_clean), d_clean))
```

```
      R2  CV_RMSE
0.735108 4.104269
```

5. Reference points

```
c(abdomen_only_R2 = summary(lm(PerBFat1 ~ Abdomen, d0))$r.squared,
  abdomen_only_CV = cv_rmse(PerBFat1 ~ Abdomen, d0),
  leak_only_CV    = cv_rmse(PerBFat1 ~ FatFreeW + Weight, d0),
  sd_outcome      = sd(d0$PerBFat1))
```

```
abdomen_only_R2 abdomen_only_CV    leak_only_CV    sd_outcome
      0.6621178      4.5789726      1.6821291      7.7508557
```

Conclusion

Excluding FatFreeW costs roughly 0.13 of R^2 and raises cross-validated error modestly. The remaining model uses only measurements that can be taken with a tape measure and a scale, which is what the study set out to do. The corrected figures are the ones reported in the README.